

PH1012 Quiz 1 (Semester 1, 2023–24) — Answers and Key Steps

Short Questions (25 points)

1. Price = **\$700,153**
 $(\$542 / \text{ft}^2) \times (3.281 \text{ ft} / 1 \text{ m})^2 \times 120 \text{ m}^2 = \$700,153$
2. $V_2 = \mathbf{0.60 \text{ m}^3}$
 $P_1 V_1 / T_1 = P_2 V_2 / T_2$ (temperatures in kelvin)
 $(1.00 \times 0.50) / (273.15 + 30) = (0.80 \times V_2) / (273.15 + 18) \Rightarrow V_2 = 0.60 \text{ m}^3$
3. $t = \mathbf{6.0 \text{ s}}$
 $x = \frac{1}{2}(v_i + v_f)t \Rightarrow t = 2x / (v_i + v_f) = 2(180) / (20.0 + 40.0) = 6.0 \text{ s}$
4. $|F| = \mathbf{23.8 \text{ N}}$
 $F_x = 65 \cos 30^\circ - 30 - 20 \sin 20^\circ = 19.45 \text{ N}$
 $F_y = 65 \sin 30^\circ - 20 \cos 20^\circ = 13.71 \text{ N}$
 $|F| = \sqrt{F_x^2 + F_y^2} = 23.8 \text{ N}$
5. **B** (the elevator's acceleration changes: speeds up, cruises, slows down)
6. **B** (pulling: the upward component of F reduces the normal force, so friction $f = \mu N$ is smaller)
7. Height at $T/4 = \mathbf{\frac{3}{4}H}$
 $H = \frac{1}{2}v_i(T/2) = \frac{1}{4}v_i T$; at $t = T/4$, speed has fallen to $v_i/2$
 $H' = \frac{1}{2}(v_i + v_i/2)(T/4) = (3/16)v_i T = \frac{3}{4}H$

Structured Questions (35 points)

1. Hot-air balloon
 - a. $U = \mathbf{11.80 \text{ N}}$, $W_A = \mathbf{9.272 \text{ N}}$, $L \approx \mathbf{2.53 \text{ N/m}^3}$
 $U = \rho_{\text{air, outside}} g = 1.204 \times 9.8 = 11.80 \text{ N (per m}^3\text{)}$
 $W_A = \rho_{\text{air, inside}} g = 0.9461 \times 9.8 = 9.272 \text{ N}$; $L = U - W_A = 2.53 \text{ N/m}^3$
 - b. $a = \mathbf{0.689 \text{ m/s}^2}$
Newton's 2nd law on the whole system (envelope air + load + skin):
 $LV - mg = (\rho_{\text{air, inside}} V + m)a$ with $V = 1800 \text{ m}^3$, $m = 320 \text{ kg} \Rightarrow a = 0.689 \text{ m/s}^2$
 - c. Air resistance (proportional to v or v^2) acts downward on the rising balloon, so the net upward force — and hence the acceleration — is smaller.
2. Overhead transmission cable
 - a. $\Delta L = \mathbf{0.14 \text{ m}}$ (total length 80.14 m also accepted)
 $\Delta L = \alpha L_0 \Delta T$ with $\Delta T = 20 - (-50) = 70^\circ \text{C} \Rightarrow \Delta L = 0.14 \text{ m}$
 - b. $M = \mathbf{24.55 \text{ kg}}$
 $M = \rho V = 2700 \times 80.4 \times \pi(0.006)^2 = 24.55 \text{ kg}$
 - c. $\Delta T = \mathbf{1.28^\circ \text{C per minute}}$
 $Q = mc\Delta T$ with $Q = 470 \times 60 = 28,200 \text{ J per minute} \Rightarrow \Delta T = 28200 / (24.55 \times 900) = 1.28^\circ \text{C}$

d. $T = 1207 \text{ N}$

Weight taken to act at the cable centre: $2T \sin \vartheta - Mg = 0$

$$\sin \vartheta = \sqrt{(80.4^2 - 80^2)}/80.4 \Rightarrow T = 1207 \text{ N}$$

e. 1. Rain / snow (extra weight on the cable). 2. Wind (extra force on the cable).

3. Block sliding off a platform

a. $v_x = 1.14 \text{ m/s}$

$$v_x = x/t = 0.48/0.42 = 1.14 \text{ m/s}$$

b. $\vartheta = 4.26^\circ$

$$-H = v_{iy}t - \frac{1}{2}gt^2: -0.90 = v_{iy}(0.42) - \frac{1}{2}(9.8)(0.42)^2 \Rightarrow v_{iy} = -0.0849 \text{ m/s}$$

$$\tan \vartheta = |v_{iy}|/v_x \Rightarrow \vartheta = 4.26^\circ$$

c. $l = 0.897 \text{ m}$

On the incline, $a = g \sin \vartheta$; speed at the base satisfies $v^2 = v_{iy}^2 + v_x^2$

$$v^2 = 0.0849^2 + 1.14^2 = 2(g \sin \vartheta)l \Rightarrow l = 0.897 \text{ m}$$